**Vidyavardhini's College of Engineering & Technology**
Department of Computer Science Engineering (Data Science)

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| **Experiment No.:** | 3 |
| **Title:** | To Create a new build job in Jenkins. |
| **Date of Performance:** | |
| **Date of Submission:** | |
| **Marks:** | |
| **Sign of Faculty:** | |

**Aim:** To Create a new build job in Jenkins.

**Theory:**

**What is a Jenkins Freestyle Project?**

Jenkins Freestyle Project is a repeatable build job, script, or pipeline that contains steps and post build actions. It is an improved job or task that can span multiple operations. It allows you to configure build triggers and offers project-based security for your Jenkins project. It also offers plugins to help you build steps and post-build actions.
The types of actions you can perform in a Jenkins build step or post-build action are quite limited. There are many standard plugins available within a Jenkins Freestyle Project to help you overcome this problem.




Fig 5.1 How to Create a Job in Jenkins

**Features of Jenkins:**

Some of the crucial features of Jenkins are the following:

- It is a free and open-source automation tool
- Jenkins provides a vast number of plugins
- It is easy to set up and install on multiple operating systems
- Provides pipeline support
- Fast release cycles
- Easy upgrades

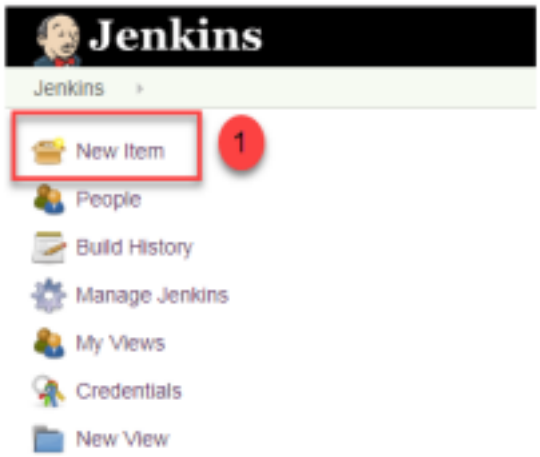
**Steps to Create a New Build Job in Jenkins:**

**Step 1: Login to Jenkins**

To create a Jenkins freestyle job, log on to your Jenkins dashboard by visiting your Jenkins installation path. Usually, it will be hosted on localhost at http://localhost:8080

**Step 2: Create New Item**

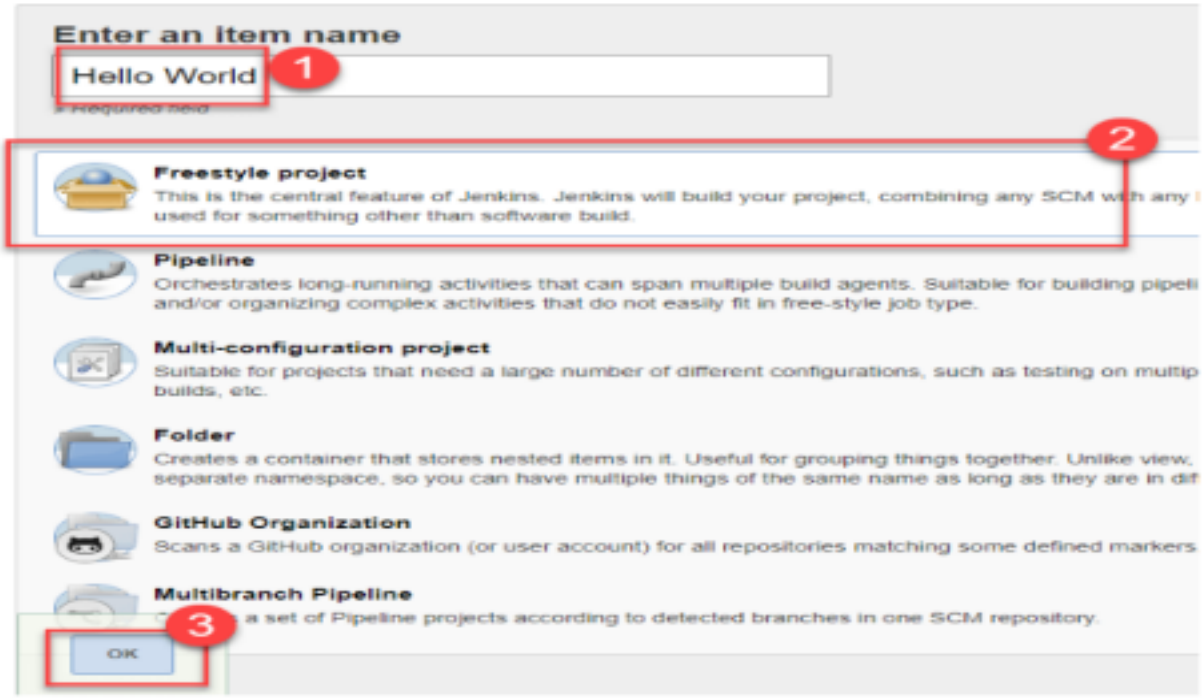
Click on "**New Item**" at the top left-hand side of your dashboard.




**Step 3: Enter Item details**

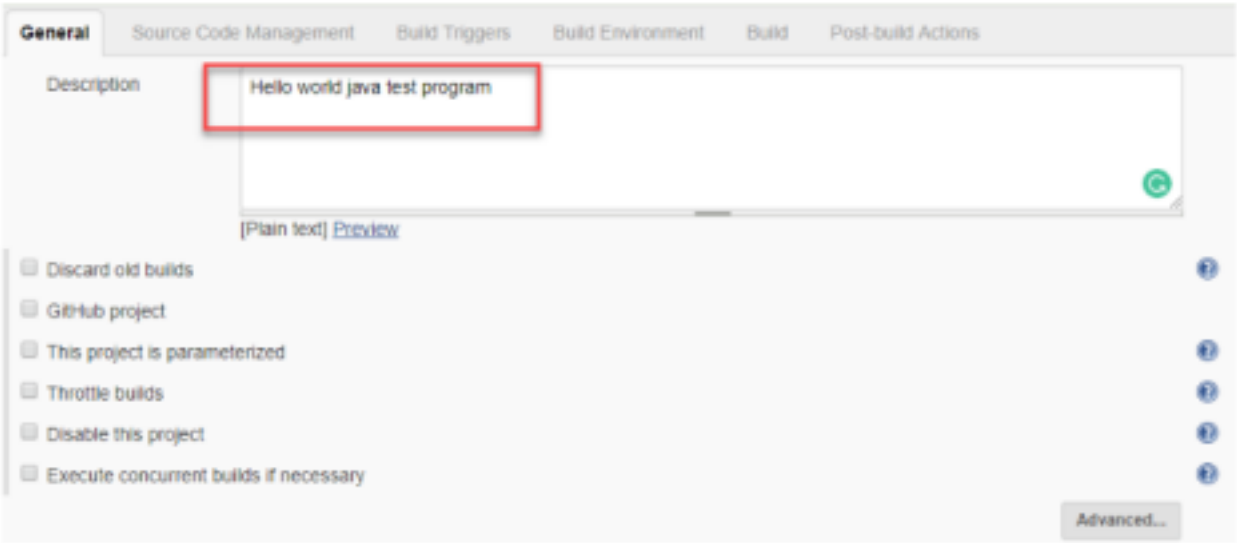
In the next screen,

1. Enter the name of the item you want to create. We shall use the "Hello world" for this demo.
2. Select Freestyle project
3. Click Okay



**Step 4: Enter Project details**

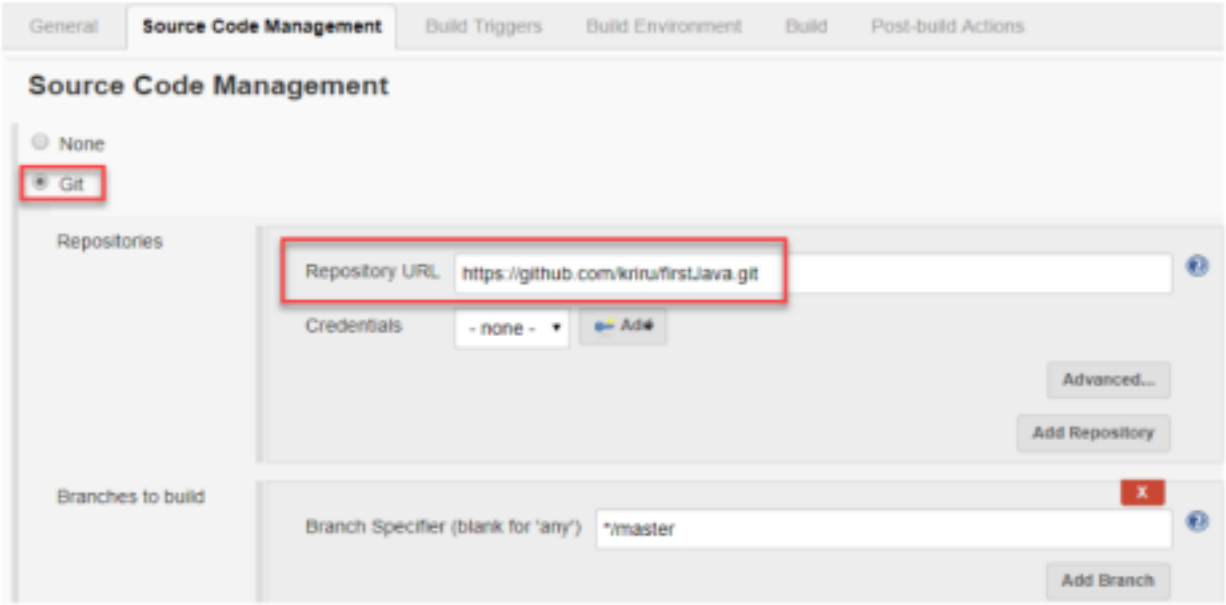
Enter the details of the project you want to test.




**Step 5: Enter repository URL**

Under Source Code Management, Enter your repository URL. We have a test repository located at https://github.com/kriru/firstJava.git

It is also possible for you to use a local repository.



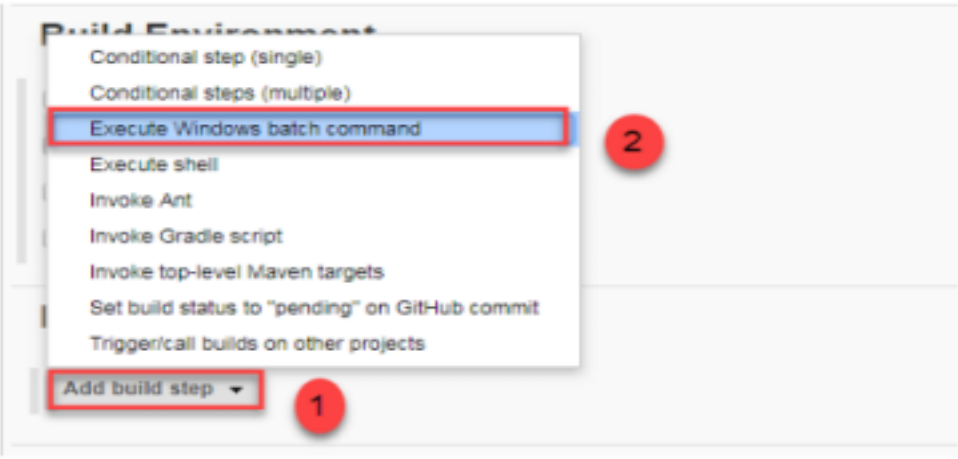

If your GitHub repository is private, Jenkins will first validate your login credentials with GitHub and only then pull the source code from your GitHub repository.

**Step 6: Tweak the settings**

Now that you have provided all the details, it's time to build the code. Tweak the settings under the **build** section to build the code at the time you want. You can even schedule the build to happen periodically, at set times.

Under **build**,

1. Click on "**Add build step**"
2. Click on "**Execute Windows batch command**" and add the commands you want to execute during the build process.




In the command window, enter the following commands and then click on the Save button. Javac HelloWorld.java

Java HelloWorld




**Step 7: Save the project**

When you have entered all the data,

1. Click **Apply**
2. **Save** the project.

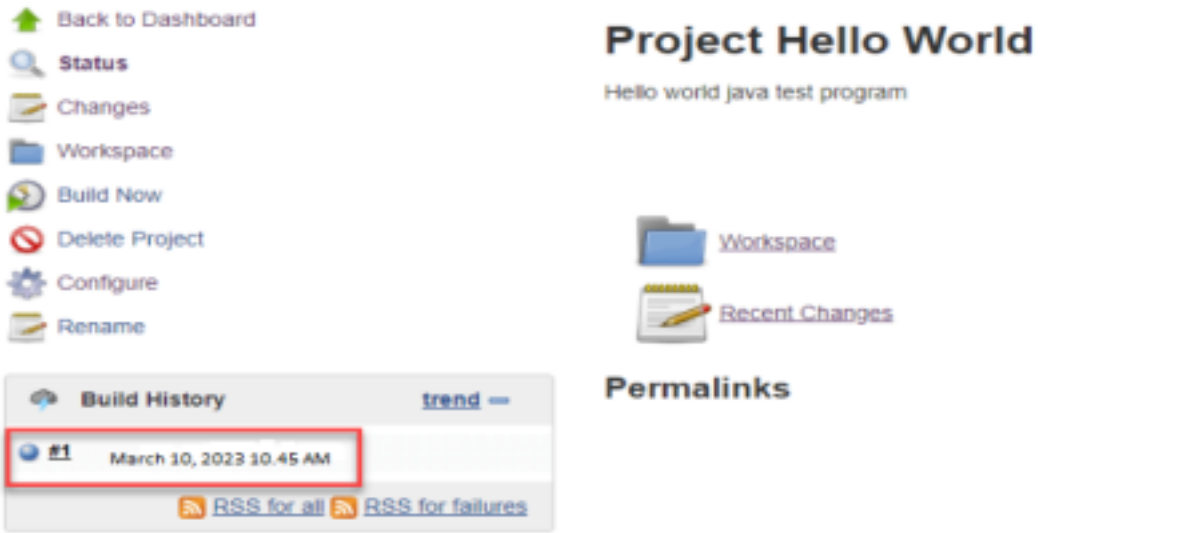
**Step 8: Build Source code**

Now, in the main screen, Click the **Build Now** button on the left-hand side to build the source code.




**Step 9: Check the status**

After clicking on **Build now,** you can see the status of the build you run under **Build History.**



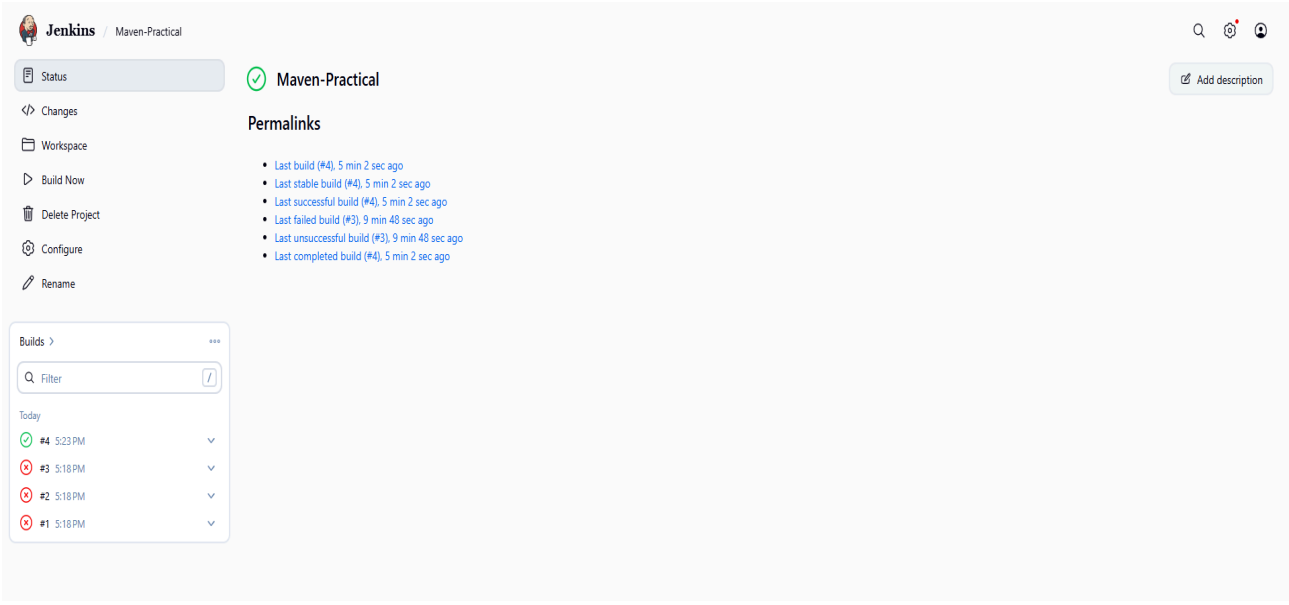
**Step 10: See the console output**

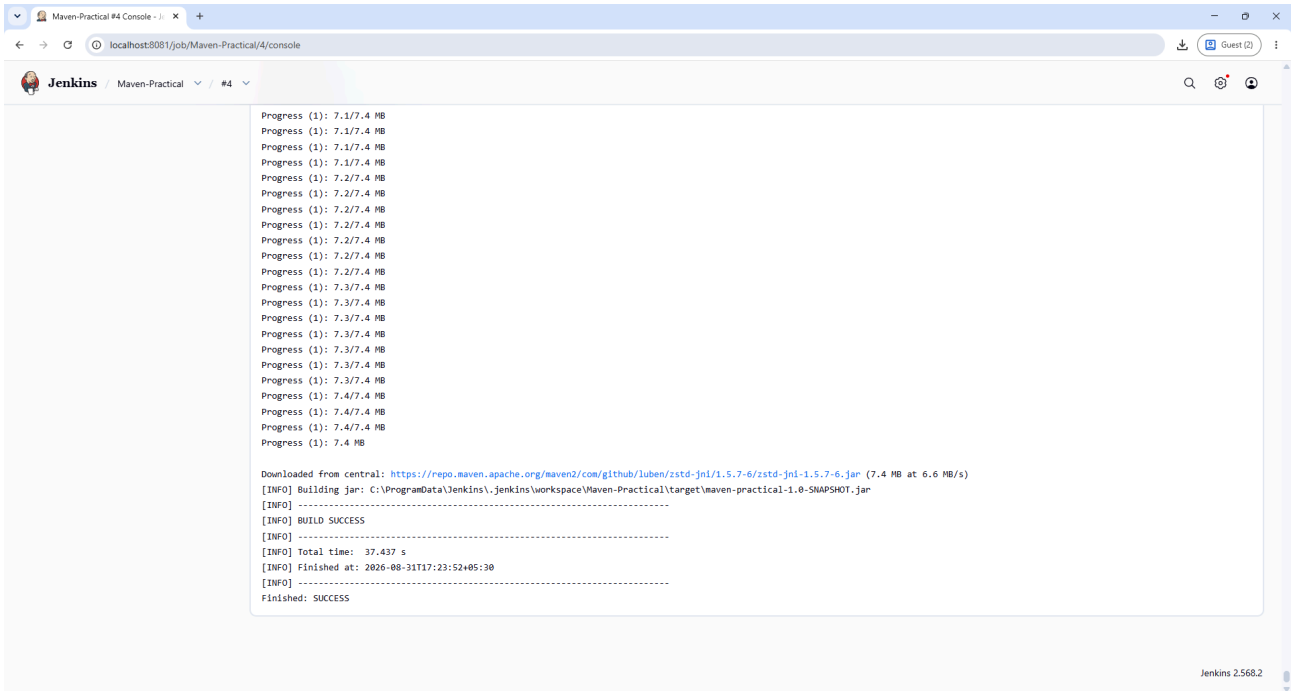
Click on the **build number** and then Click on **console output** to see the status of the build you run. It should show you a success message.

Out




Output -






## Conclusion:

1. Which SCM tools Jenkins supports?

Ans - Jenkins supports the following Source Code Management (SCM) tools:

1. Git – Distributed version control system.
2. Subversion (SVN) – Centralized version control system.
3. CVS – Concurrent Versions System.
4. Mercurial – Distributed version control system.
5. Perforce – Version control system used for managing source code.

2. What are the various ways in which build can be scheduled in Jenkins?

Ans - Jenkins provides the following ways to schedule a build:

1. Trigger builds remotely: A build is triggered using a remote URL or API.
2. Build periodically: Runs a build at a specified time using cron syntax.
3. Poll SCM: Checks the source code repository at regular intervals and starts a build when changes are detected.
4. Build after other projects are built: Starts a build after another specified project completes.
5. GitHub webhook: Automatically triggers a build when changes are pushed to GitHub.
6. Manual build: A user starts the build by clicking Build Now.